A FINITELY GENERATED IDEAL CONTAINED IN THE RADICAL OF I HAS A POWER CONTAINED IN I

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the frozen snapshot stages the chapter's own preamble file beside it (preamble.tex, 217 lines, 8116 bytes, the same shared Stacks-wide preamble the published chow and etale-cohomology cards use); the first line of the pinned chapter file is \input{preamble}, so no preamble text lives inside the chapter itself. The card takes that file's lines 8-18 (the verbatim package and the three \comment-based environments reference/slogan/history at lines 9-12, section 13.2, together with the Xy-pic setup) and its lines 154 to the runtime end of the file (the native theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointers are located at their \label lines inside the pinned chapter file; the cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The counted statement contains a slogan environment, which the chapter's own preamble defines as a comment environment at lines 11-12; the preamble spans therefore run 8-18 and cover it.

Editorial note. The counted spans contain no \ref and no \cite, so the card carries no navigation stub and no bibliography.

Allowed standard prerequisites (editorial).

- Expansion of a power of a sum of s elements. Standard multinomial expansion, used at the last step of the counted proof (line 6532). Shows that every monomial of degree n = d_1 + ... + d_s + 1 in the generators f_i has one exponent at least d_i, hence lies in I.
- Membership in the radical, read generator by generator. The hypothesis J contained in the radical of I, applied at proof line 6531. Supplies, for each of the finitely many generators f_i of J, an exponent d_i with f_i^{d_i} in I.

Lemma 0.1. *Let R be a Noetherian ring. Let I, J be ideals of R. Suppose $J \subset \sqrt{I}$. Then $J^n \subset I$ for some n. In particular, in a Noetherian ring the notions of "locally nilpotent ideal" and "nilpotent ideal" coincide.*

Proof. Say $J = (f_1, \ldots, f_s)$. By assumption $f_i^{d_i} \in I$. Take $n = d_1 + d_2 + \ldots + d_s + 1$. □

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